TUTORIAL - 7 (KNN)

Training Dataset:

Movie ID	Action Scenes	Romance Scenes	Happy Ending	Genre
	15	2	0	A
	1	12	1	R
	18	1	0	A
	2	14	1	R
	14	3	1	A

Test Data:

Action Scenes	Romance Scenes	Happy Ending	Actual Genre
10	5	1	A

calculate dist b/w $(10, 5, 1) - (15, 2, 0)$

Euclidean distance

$(15, 2, 0) \rightarrow A$

$$d_1 = \sqrt{(x_{test1} - x_1)^2 + (x_{test2} - x_2)^2 + (x_{test3} - x_3)^2}$$

$$= \sqrt{(10-15)^2 + (5-2)^2 + (1-0)^2}$$

$$= \sqrt{25 + 9 + 1}$$

$$= \sqrt{35} = 5.92 \rightarrow A, \; w_1 = \frac{1}{5.92} = 0.19$$

$(1, 12, 1)$

$$d_2 = \sqrt{(10-1)^2 + (5-12)^2 + (1-1)^2}$$

$$= \sqrt{81 + 49 + 0}$$

$$= \sqrt{130} = 11.40 \rightarrow R \rightarrow w_2 = 0.08$$

$(18, 1, 0) \rightarrow A$

$$d_3 = \sqrt{(10-18)^2 + (5-1)^2 + (1-0)^2}$$

$$= \sqrt{64 + 16 + 1}$$

$$= \sqrt{81} = 9 \rightarrow A \rightarrow w_3 = 0.11$$

$(2, 14, 1)$ $d_4 = 12.04 \rightarrow R \rightarrow w_4 = 0.08$

$(14, 3, 1) \rightarrow A$ $d_5 = 4.47 \rightarrow A \rightarrow w_5 = 0.22$

Scanned with OKEN Scanner

$K = 5$

Sense weight $= \frac{1}{d}$

$w = \frac{1}{d}$

weights of A $= w_1 + w_3 + w_5$

$= 0.19 + 0.11 + 0.22$

$= 0.52 > 0.16$

weights of R $= w_2 + w_4$

$= 0.08 + 0.08$

$= 0.16$

$\therefore$ More is Action.

Regression:

Age (x_1)	Previous value (x_2) (in Lakh)	car price (y) in lakh
2	3	2
3	4	3
5	1	0.5

Test: $(4, 2, ?)$ $\hat{y} = 1.75$

$= 5 \cdot Y$

$(4, 2)$

$(2, 3) - d_1 = \sqrt{(4-2)^2 + (2-3)^2} = \sqrt{4+1} = \sqrt{5} = 2.24$

$(3, 4) - d_2 = \sqrt{(4-3)^2 + (2-4)^2} = 2.24 \rightarrow 3$

$(5, 1) - d_3 = \sqrt{(4-5)^2 + (2-1)^2} = \sqrt{2} = 1.41$ shortest 0.5

If $K = 2$

$\hat{y} = \frac{1}{K} \sum_{i=1}^{K} y_i = \frac{3 + 0.5}{2} = 1.75$